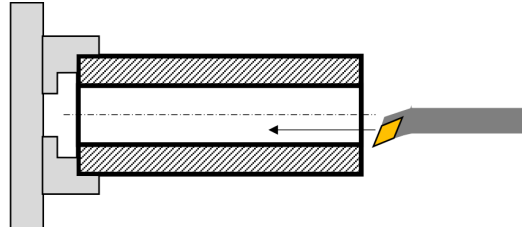


CAM – SECOND MIDTERM EXAM 24/01/2022

SOLUTIONS TO NUMERICAL PROBLEMS

TURNING

The machining of the internal cylinder of a tubular component is carried out over the entire length of the workpiece with the set-up shown in the figure.



Data: External diameter = 250 mm, initial internal diameter = 152 mm, final internal diameter = 155 mm, total length of the workpiece = 300 mm; Kronenberg parameters are $k_{c0,4} = 2500$ MPa, $x = 0,29$; tool changing time = 90 s, tool radius = 0,4 mm, main entering angle $k_r = 120^\circ$, secondary entering angle $k'_r = 30^\circ$, roughness requirement $R_a = 0,8 \mu\text{m}$, feed $f = 0,1$ mm/rev; machine tool nominal power 5 kW, machine tool maximum velocity 3000 RPM.

Please:

- 1) Verify if the roughness requirement is met.
- 2) Considering that a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 100 mm^2 and friction coefficient equal to 0,3, determine the minimum clamping pressure to properly hold the workpiece.
- 3) Determine the cutting power and the cutting time in the hypothesis to consider a cutting speed that maximizes the production rate. Is the operation feasible with the considered machine tool? Please, consider a real cutting force equal to 600 N. It is also known that if the cutting speed is equal to 310 m/min the tool life is equal to 5 minutes, and if the cutting speed is equal to 250 m/min the tool life is equal to 15 minutes.

Solution

- 1) *Verify if the roughness requirement is met*

Considering the Schmalz's equation (tool nose radius greater than zero), the expected roughness is

$$R_a = \frac{f^2 \cdot 10^3}{32 \cdot r} = \frac{0,1^2 \cdot 10^3}{32 \cdot 0,4} = 0,78 \mu\text{m} \leq 0,8 \mu\text{m}$$

The roughness requirement is met.

We need to check if the Schmalz's equation hypotheses hold:

$$\begin{cases} \frac{f}{2} \leq r \sin k_r \\ \frac{f}{2} \leq r \sin k'_r \end{cases} \quad \begin{cases} f \leq 2r \sin k_r \\ f \leq 2r \sin k'_r \end{cases} \quad \begin{cases} 0,1 \leq 2 \cdot 0,4 \cdot \sin 120^\circ = 0,693 \rightarrow ok \\ 0,1 \leq 2 \cdot 0,4 \cdot \sin 30^\circ = 0,4 \rightarrow ok \end{cases}$$

- 2) *Considering that a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 100 mm^2 and friction coefficient equal to 0,3, determine the minimum clamping pressure to properly hold the workpiece*

To check if the workpiece is properly held, the resistant torque should be greater than the cutting torque.

$$T_c = F_c \cdot \frac{D_f/2}{1000} \leq T_r = z \cdot \mu \cdot p \cdot A \cdot \frac{D_{clamping}/2}{1000}$$

Therefore, we need to compute the cutting force, as follows:

$$k_c = k_{c,0,4} \cdot \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2500 \cdot \left(\frac{0,4}{0,1 \cdot \sin 120^\circ} \right)^{0,29} = 3896 \text{ MPa}$$

$$a_p = \frac{D_f - D_i}{2} = \frac{155 - 152}{2} = 1,5 \text{ mm}$$

$$A_D = f \cdot a_p = 0,1 \cdot 1,5 = 0,15 \text{ mm}^2$$

$$F_c = k_c \cdot A_D = 3896 \cdot 0,15 = 584,4 \text{ N}$$

Consequently, the following minimum clamping pressure is needed:

$$p \geq p_{\min} = \frac{F_c \cdot \frac{D_f/2}{1000}}{z \cdot \mu \cdot A \cdot \frac{D_{\text{clamping}}/2}{1000}} = \frac{F_c \cdot D_f}{z \cdot \mu \cdot A \cdot D_{\text{clamping}}} = \frac{584,4 \cdot 155}{3 \cdot 0,3 \cdot 100 \cdot 250} = 4,03 \text{ MPa}$$

- 3) Determine the cutting power and the cutting time in the hypothesis to consider a cutting speed that maximizes the production rate. Is the operation feasible with the considered machine tool? Please, consider a real cutting force equal to 600 N. It is also known that if the cutting speed is equal to 310 m/min the tool life is equal to 5 minutes, and if the cutting speed is equal to 250 m/min the tool life is equal to 15 minutes.

To maximize the production rate, we need to minimize the production time. Therefore,

$$T_{\text{opt}} = T_t \left(\frac{1}{n} - 1 \right)$$

We need to determine the Taylor's parameters, as follows

$$n = \ln \left(\frac{v_{c1}}{v_{c2}} \right) / \ln \left(\frac{T_2}{T_1} \right) = \ln \left(\frac{310}{250} \right) / \ln \left(\frac{15}{5} \right) = 0,196$$

$$C = v_c \cdot T^n = 310 \cdot 5^{0,196} = 425$$

$$T_{\text{opt}} = T_t \left(\frac{1}{n} - 1 \right) = \frac{90}{60} \left(\frac{1}{0,196} - 1 \right) = 6,153 \text{ min}$$

and

$$v_{c,\text{opt}} = \frac{C}{T_{\text{opt}}^n} = \frac{425}{6,153^{0,196}} = 298 \frac{\text{m}}{\text{min.}}$$

Therefore,

$$P_c = \frac{F_c \cdot v_c}{60} = \frac{600 \cdot 298}{60} = 2.980 \text{ W} = 2,98 \text{ kW} < 5 \text{ kW}$$

Moreover, the revolutions per minute are:

$$m = \frac{1000 \cdot v_c}{\pi \cdot D_f} = \frac{1000 \cdot 298}{\pi \cdot 155} = 612 \text{ RPM} < m_{\max} = 3000 \text{ RPM}$$

It follows that the machining operation is feasible.

The contact time can be determined as follows:

$$t_c = \frac{L_c}{v_f} = \frac{L_c}{f \cdot m} = \frac{300}{0,1 \cdot 612} = 4,9 \text{ min.}$$

REAMING

Determine the cutting power and the cutting time needed to finish a through hole with a reaming operation.

The final diameter of the finished hole is equal to 45 mm and its height is equal to 60 mm. The workpiece material is a steel and, considering the cutting conditions, the cutting pressure is equal to 2600 MPa. The chosen tool has a diameter of 45 mm, 8 straight cutting edges, and the suggested cutting parameters are: depth of cut equal to 0,3 mm, feed per tooth equal to 0,16 mm/rev., and cutting speed equal to 90 m/min.

Solution

1) Determine the cutting power

In the reaming operation each cutting edge will remove material with a depth of cut equal to

$$a_p = \frac{D_{final} - D_{initial}}{2} = 0,3 \text{ mm}$$

It follows that the initial diameter of the hole must be equal to

$$D_{initial} = D_{final} - 2a_p = 45 - 2 \cdot 0,3 = 44,4 \text{ mm}$$

The chip area before deformation is equal to

$$A_D = f_z \cdot a_p = 0,16 \cdot 0,3 = 0,048 \text{ mm}^2$$

and the cutting force per each cutting edge is equal to

$$F_c = k_c A_D = 2600 \cdot 0,048 = 124,8 \text{ N}$$

In a reaming operation all the cutting edges are simultaneously working with identical cutting parameters, therefore

$$T_c = Z \cdot F_c \cdot \frac{D_{final} + D_{initial}}{4} \cdot \frac{1}{1000} = 8 \cdot 124,8 \cdot \frac{45 + 44,4}{4} \cdot \frac{1}{1000} = 22,3 \text{ Nm}$$

Therefore, the cutting power is equal to

$$P_c = T_c \omega$$

The angular velocity is the following

$$n = \frac{1000 \cdot v_c}{\pi D_{final}} = \frac{1000 \cdot 90}{\pi \cdot 45} = 637 \text{ RPM}$$
$$\omega = \frac{2\pi \cdot n}{60} = \frac{2\pi \cdot 637}{60} = 66,7 \frac{\text{rad}}{\text{s}}$$

Therefore:

$$P_c = T_c \omega = 22,3 \cdot 66,7 = 1.487 \text{ W} = 1,5 \text{ kW}$$

2) Determine the cutting time

The cutting time is equal to

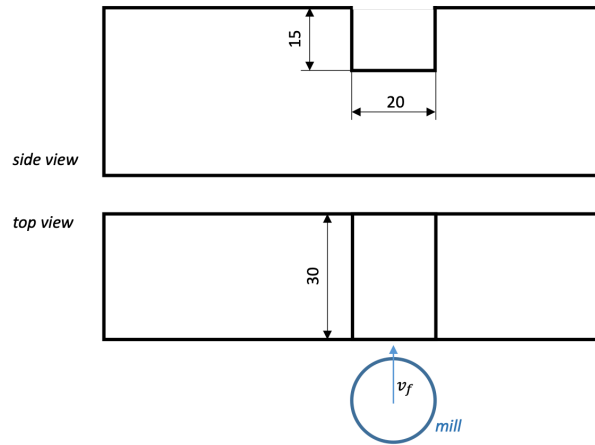
$$t_c = \frac{L_c}{v_f} = \frac{H}{Z \cdot f_z \cdot n}$$

Therefore:

$$t_c = \frac{60}{8 \cdot 0,16 \cdot 637} \cdot 60 = 4,4 \text{ s}$$

MILLING

A slot has to be machined from a titanium alloy prismatic block as in the figure.



Considering an end milling operation, determine:

- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 10 kW and an efficiency of 0,9.
- 2) The cutting time needed to carry out the operation and the material removal rate.

Data: number of cutting edges $Z = 4$, entering angle = 90° , revolutions per minute equal to 600 RPM, feed velocity equal to 80 mm/min., cutting pressure equal to 2500 MPa.

Solution

- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 10 kW and an efficiency of 0,9.

Considering the end milling, a particular kind of face milling operation, the tool has a diameter equal to the slot width 20 mm. Considering the geometry of the cutting, the engagement angle is equal to 180° and the angular step between two consecutive cutting teeth is equal to $\varphi_0 = \frac{2\pi}{Z} = \frac{360}{4} = 90^\circ$, then the number of simultaneously working teeth is equal to:

$$z = \frac{\varphi}{\varphi_0} = \frac{180^\circ}{90^\circ} = 2$$

Therefore, we must consider the worst-case scenario when two cutting edges are simultaneously cutting, that we have when the angular position of the two cutting edge is symmetric with respect to the feed direction, so:

- Cutting tooth 1, angular position: $\theta_1 = \frac{\varphi_0}{2} = \frac{90^\circ}{2} = 45^\circ$
- Cutting tooth 2, angular position: $\theta_2 = -\frac{\varphi_0}{2} = -\frac{90^\circ}{2} = -45^\circ$

To answer the question, we notice that:

$$h_D(\theta_1) = f_Z \cos \theta_1 = f_Z \cdot \cos \frac{\varphi_0}{2}$$

$$h_D(\theta_2) = f_Z \cos \theta_2 = f_Z \cdot \cos -\frac{\varphi_0}{2} = f_Z \cdot \cos \frac{\varphi_0}{2}$$

So:

$$h_D(\theta_1) = h_D(\theta_2) = h_D$$

Moreover, being $v_f = 80 \frac{\text{mm}}{\text{min}}$

$$v_f = Z \cdot f_Z \cdot n \rightarrow f_Z = \frac{v_f}{Z \cdot n} = \frac{80}{4 \cdot 600} = 0,033 \frac{\text{mm}}{\text{rev}}$$

Therefore:

$$h_D = f_Z \cdot \cos \frac{\varphi_0}{2} = 0,033 \cdot \cos 45^\circ = 0,023 \text{ mm}$$

$$A_D = h_D \cdot a_p = 0,023 \cdot 15 = 0,345 \text{ mm}^2$$

$$F_c = k_c A_D = 2.500 \cdot 0,345 = 862,5 \text{ N}$$

$$T_c = F_{c,\theta_1} \frac{D/2}{1000} + F_{c,\theta_2} \frac{D/2}{1000} = 2 \cdot F_c \frac{D/2}{1000} = 2 \cdot 862,5 \frac{20/2}{1000} = 17,25 \text{ Nm}$$

Being $n = 600$ RPM, it follows:

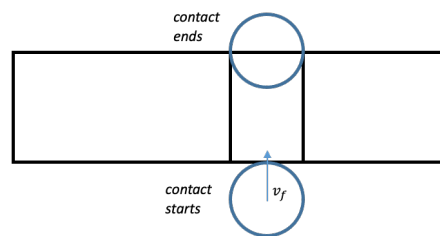
$$\omega = \frac{2\pi \cdot n}{60} = \frac{2\pi \cdot 600}{60} = 62,83 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 17,25 \cdot 62,83 = 1.084 \text{ W}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 10 = 9 \text{ kW}$$

Therefore, being $P_c < P_{c,limit}$, the milling operation is feasible.

2) The cutting time needed to carry out the operation and the material removal rate



Referring to the figure, the cutting time is equal to:

$$t_c = \frac{L_{workpiece} + \frac{D}{2}}{v_f} = \frac{30 + \frac{20}{2}}{80} = 0,5 \text{ min.} = 30 \text{ s}$$

The material removal rate is the volume of material removed per unit of time:

$$MRR = \frac{V}{t_c} = \frac{30 \cdot 20 \cdot 15}{30} = \frac{9000}{30} = 300 \frac{\text{mm}^3}{\text{s}}$$